SPECIFICATIONS

ELECTRICAL CHARACTERISTICS

AV_{DD} = 5.0 V ± 5%, DV_{DD} = 1.8 V ± 5%, 1.1 V ≤ V_{LOGIC} ≤ 1.9 V, V_{REF} = 2.5 V, −40°C ≤ T_A ≤ +105°C, output amplifier AD8675, unless otherwise noted.

Table 2.

Parameter[1]	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments
STATIC PERFORMANCE						
Resolution		16			Bits	
Relative Accuracy (INL)		−2		+2	LSB	5 V range only
		−4		+4	LSB	All other ranges[2]
Differential Nonlinearity (DNL)		−1		+1	LSB	Precision mode: −40°C to +105°C, fast mode: 0°C to 85°C
		−2		+2	LSB	Fast mode: −40°C to +105°C
		−2		+2	LSB	0 V to 2.5 V range, fast or precision modes[2]
Offset Error			0.03		%FSR	Midscale, 25°C
Offset Error Drift[2]			2	8	ppm FSR/°C	0 V to 5 V and 0 V to 10 V ranges
			4	16	ppm FSR/°C	All other ranges
Full-Scale Error			0.04		%FSR	25°C
Full-Scale Error Drift[2]			1	5	ppm FSR/°C	0 V to 5 V and 0 V to 10 V ranges
			4	12	ppm FSR/°C	All other ranges
Zero-Scale Error[3]			0.05		%FSR	25°C
Zero-Scale Error Drift[2]			3.5	8	ppm FSR/°C	0 V to 5 V and 0 V to 10 V ranges
			7	16	ppm FSR/°C	All other ranges
Total Unadjusted Error (TUE)		−0.5		+0.5	%FSR	
DC Power Supply Rejection Ratio (PSRR)			0.6		mV/V	DAC code = midscale
OUTPUT CHARACTERISTICS						
Output Current	$I_{OUT}X$		1.6		mA	Absolute value
REFERENCE OUTPUT						
Output Voltage		2.492	2.5	2.508	V	At 25°C, over lifetime
Voltage Reference Temperature Coefficient (TC)[4]			3	10	ppm/°C	
Output Impedance			50		mΩ	
Output Voltage Noise			2.7		μV rms	0.1 Hz to 10 Hz
Output Voltage Noise Density			173		$nV/\sqrt{Hz}$	f = 1 kHz, no load on V_{REF}
			164		$nV/\sqrt{Hz}$	f = 10 kHz, no load on V_{REF}
Capacitive Load Stability[2]				10	μF	
Load Regulation			50		μV/mA	At 25°C
Output Current Load Capability			±8		mA	
Line Regulation			135		μV/V	At 25°C
REFERENCE INPUT						
Reference Current			1		μA	
Reference Input Range[2]	V_{REF}	2.4	2.5	2.6	V	
Reference Input Impedance			3		MΩ	
LOGIC INPUTS						
Input Current	I_I	−1		+1	μA	Per pin
Input Low Voltage	V_{IL}			0.35 × V_{LOGIC}	V	
Input High Voltage	V_{IH}	0.65 × V_{LOGIC}			V	
Pin Capacitance	C_I		4		pF	

SPECIFICATIONS

Table 2. (Continued)

Parameter[1]	Symbol	Min	Typ	Max	Unit	Test Conditions/Comments
LOGIC OUTPUTS						
Output Low Voltage	V_{OL}			0.20 × V_{LOGIC}	V	I_{SINK} = 100 µA
Output High Voltage	V_{OH}	0.80 × V_{LOGIC}			V	I_{SOURCE} = 100 µA
Pin Capacitance	C_O		4		pF	
POWER REQUIREMENTS						
V_{LOGIC} Pin		1.1	1.8	1.89	V	
V_{LOGIC} Current	I_{LOGIC}		1	7.5	µA	$V_{IH} = V_{LOGIC} \times 0.9$, $V_{IL} = V_{LOGIC} \times 0.1$
V_{LOGIC} Dynamic Current	$I_{LOGIC\_DYNAMIC}$		3	5	mA	SCLK = 66 MHz, quad SPI DDR, $V_{IH} = V_{LOGIC} \times 0.65$, $V_{IL} = V_{LOGIC} \times 0.35$
DV_{DD} Pin		1.71	1.8	1.89	V	
DV_{DD} Current	I_{DVDD}		0.5	0.8	mA	
DV_{DD} Dynamic Current	$I_{DVDD\_DYNAMIC}$		33	40	mA	SCLK = 66 MHz, quad SPI DDR
AV_{DD} Pin		4.75	5	5.25	V	
AV_{DD} Current	I_{DD}		12	15	mA	Channel 0 zero-scale, 0 V to ±5 V range
AV_{DD} Power-Down Current	I_{DD}		0.6		mA	After reset, DACs powered down
AV_{DD} Reset Current	I_{DD}		120		µA	$\overline{\text{RESET}}$ asserted

1 See the Terminology section.
2 Guaranteed by design and characterization, not production tested.
3 Measured at zero code.
4 Reference temperature coefficient is calculated as per the box method.

AC CHARACTERISTICS

AV_{DD} = 5.0 V ± 5%, DV_{DD} = 1.8 V ± 5%, 1.1 V ≤ V_{LOGIC} ≤ 1.9 V, −40°C ≤ T_A ≤ +105°C, measured with the ADA4807-1 external amplifier, unless otherwise noted.

Table 3.

Parameter[1]	Min	Typ	Max	Unit	Test Conditions/Comments
DYNAMIC PERFORMANCE					
Output Voltage Settling Time		100		ns	2 V step, 0.1% error, 0 V to 5 V range
		75		ns	2 V step, 1% error, 0 V to 5 V range
		65		ns	60 mV step, 0.1% error, 0 V to 5 V range
		15		ns	60 mV step, 1% error, 0 V to 5 V range
Slew Rate		100		V/µs	Full-scale step, 0 V to 2.5 V range
Digital-to-Analog Glitch Impulse		50		pV×s	0 V to 5 V range, ±1 LSB change around major carry
Digital Feedthrough		25		pV×s	50 MHz clock, $R_{FB}2\_x$
AC PSRR		80		dB	1 kHz, $R_{FB}1\_x$
		43		dB	1 MHz, $R_{FB}1\_x$
Output Noise Spectral Density		15		nV/$\sqrt{\text{Hz}}$	DAC code = midscale, external reference, 10 kHz, NCAPx = 1.2 µF, PCAPx = none, $R_{FB}1\_x$
		30		nV/$\sqrt{\text{Hz}}$	$R_{FB}2\_x$
		60		nV/$\sqrt{\text{Hz}}$	$R_{FB}4\_x$
Output Noise		3.8		µV_{RMS}	DAC code = midscale, external reference, 1 Hz to 10 kHz, NCAPx = 1.2 µF, PCAPx = none, $R_{FB}1\_x$
		7.6		µV_{RMS}	$R_{FB}2\_x$
		15.4		µV_{RMS}	$R_{FB}4\_x$
Total Harmonic Distortion (THD)		−105		dB	0 V to 5 V range, f_{OUT} = 1 kHz